THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

May 8, 2023 TIME: 2:30 – 5:30 PM

Answer ALL questions. Section A carries 33 marks, and Section B carries 67 marks.

Use a separate answer book for each section.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (11 multiple choice questions with 3 marks each)

You may need to use Appendix A at the end of the exam paper for your calculations.

1. What is the volume of 1 mole of ideal gas law at a high-altitude region? Assuming the air pressure is 0.64 atm and the temperature is 10 °C.
   (a) 24.5 L; (b) 35.0 L; (c) 36.3 L; or (d) 38.3 L

2. How much energy should be input to heat 1 bottle of water (750 mL) from 20 °C to 100 °C?
   (a) 250.8 kJ; (b) 313.5 kJ; (c) 334.4 kJ; or (d) 418 kJ.

3. Which of the following light possess the highest energy?
   (a) Infrared; (b) Blue; (c) Green; or (d) Ultraviolet.

4. Which of the following disinfectant has the strongest disinfection power?
   (a) free chlorine; (b)chloramine; (c) chlorine dioxide; or (d) ozone.

5. A circular primary sedimentation tank receives a wastewater influent of 24,000 m^3/d. The designed depth of the tank is 5 m while the hydraulic retention time (HRT) is 2 hours. What is the designed surface overflow rate for the tank?
   (a) 1.0 m/h; (b) 1.5 m/h; (c) 2.5 m/h; or (d) 3.0 m/h.

6. Which is **NOT** the key factor affecting the food over microorganism ratio in activated sludge process?
   (a) influent TSS; (b) influent BOD; (c) influent flow rate; or (d) biomass concentration.

7. Which is **NOT** the advantage of anaerobic digestion over aerobic processes?
   (a) Less energy required; (b) Low production of waste biological sludge; (c) Larger reactor volume required; or (d) Low nutrient required.

8. Which of the following statements regarding dissolved oxygen (DO) is **NOT** true?
   (a) Aquatic life need DO to survive;
   (b) Bacteria uses DO to decompose organic matters;
   (c) As DO levels fall, undesirable odors, tastes, and colors come down; or
   (d) Higher salinity leads to lower DO.

9. A sample of water contains various ions including 40.0 mg/L Ca^{2+}, 12.2 mg/L Mg^{2+}, 7.0 mg/L K^+, 110.0 mg/L HCO_3^-, 67.2 mg/L SO_4^{2-}, 11.0 mg/L Cl^-, and certain amount of Na^+. Assuming no other constituents are missing, what is the total hardness of this water sample in mg/L as $CaCO_3$?
   (a) 50 mg/L as $CaCO_3$; (b) 100 mg/L as $CaCO_3$; (c) 150 mg/L as $CaCO_3$; or (d) 200 mg/L as $CaCO_3$.

10. Which of the following is **NOT** the production of catalytic converter?
    (a) CO_2; (b) H_2O; (c) N_2; (d) SO_2; (e) none of the above.

11. Which of the following is primary pollutants?
    (a) SO_3; (b) O_3; (c) CH_4; (d) N_2; (e) none of the above.

Section B. (Three questions with unequal marks)

You may need to use Appendix A at the end of the exam paper for your calculations.

12. The equivalent partial pressure of CO_2 of groundwater is 0.01 atm. The aquifer contains limestone so that the groundwater has reached equilibrium with $CaCO_3$ (s). The pH of the groundwater is 7.5.

    a) Determine $[H_2CO_3]$, $[HCO_3^-]$, $[CO_3^{2-}]$, and $[Ca^{2+}]$; (10 marks)
    b) Based on your answers for (a), perform a charge balance for the groundwater. Is the charge balance satisfied? Provide brief explanations of your findings (**in no more than 100 words**). (12 marks)

13. (a) What are **three similarities** and **three differences** between water treatment and wastewater treatment.

    (6 marks)

    (b) Draw the oxygen sag curve and briefly introduce factors that affect the curve (< **200 words**, **up to 1 figure** is allowed).

    (5 marks)

    (c) An inland city has an available water source with the hardness of 600 mg/L as $CaCO_3$. If the water is used for drinking purposes, please design an appropriate treatment process, and explain the design purposes.

    (12 marks)

14. (a) A circular, well-mixed aeration biological reactor is 1.9 m deep and 20 m in diameter. It receives 400 m^3/d of wastewater with a BOD_5 of 336 mg/L for treatment. The biodegradation rate constants for the wastewater are: endogenous decay constant $k_d = 0.08$ d^{-1}, maximum specific microbial growth rate $\mu_m = 1.10$ d^{-1}, and the half velocity constant $K_s = 76.0$ mg BOD_5/L. What is the efficiency of the biological reactor?

    (12 marks)

    (b) Please respectively discuss the formation and effects of ocean acidification and rising sea level (< **300 words**)

    (10 marks)

Appendix A Information for Reference

I.
Atomic weights: H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

II.
Properties of water:
Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

III.
Ideal gas law constant = 0.082056 L·atm/(K·mol)
Partial pressure of CO_2 in atmosphere: $1 \times 10^{-3.5}$ atm
Henry's Law constant for CO_2: 0.03162 mol/(L·atm)
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Solubility of $CaCO_3$(s): $pK_{so} = 8.3$
Henry's Law constant for O_2 (25 °C) = 0.001263 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1 \times 10^{-14}$
Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq
EW for Mg^{2+} = 12.2 mg/meq
EW for $CaCO_3$ = 50 mg/meq

- END OF PAPER -